The Paleomagnetism of the Aden Volcanics

E. IRVING AND D. H. TARLING

Department of Geophysics, Australian National University
Canberra, A.C.T., Australia

Abstract. The directions and intensities of the natural remanent magnetization of 164 specimens from twelve sites in the Aden Volcanics have been measured. The stability of these directions in strong alternating fields is demonstrated. The results, besides providing the first information about the secular variation and reversals of the earth's magnetic field in low latitudes during the late Cenozoic, were also found to be of interest in connection with the origin of the Red Sea. The directions deviate systematically from true north, the average declination being N 7° W. This deviation could be due to an anticlockwise rotation of Arabia, relative to Africa, by an amount sufficient to account for the formation of the Red Sea. However, explanations other than rotation are possible, and these cannot be tested until results are available from northeast Africa.

1. INTRODUCTION

Paleomagnetic studies of Upper Tertiary and Quaternary rocks have so far been confined to high or intermediate latitudes, and no information is available from low latitudes on such questions as the supposed average dipole nature of the earth's field, its secular variation, and its possible reversal during this period of time. Information on the secular variation in low latitudes is likely to be significant for the models used in the analysis of paleomagnetic results. This paper contains a description of some magnetic properties of the volcanic rocks of Aden (12°47'N, 45°01'E). These results relate to geomagnetism and also to the tectonic problem of the origin of the Red Sea.

The main faulting associated with the Red Sea was initiated in Middle Tertiary times. A depression was present during the Oligocene which deepened during the Miocene. It is likely that this depression represented an initial stage and that the Red Sea as it is today was formed later. Without attempting a comprehensive survey of the published discussions of this problem, we find it convenient to recognize three possible modes of formation.

A. Compressional rifting; under this hypothesis it is assumed that the Red Sea was a segment of the crust, bounded by reversed faults, which has been pushed down.

B. Tensional rifting; in this case it is assumed that the Red Sea is a sunken crustal segment, bounded by normal faults.

C. Dilatation of tensional faults; it is here supposed that the crustal blocks of Arabia and Africa moved apart, leaving between them subcrustal material, which, because of its greater density, remained at a lower level.

Hypotheses A and B imply that Africa and Arabia have moved only small distances (order of 1 km) relative to each other, since the faulting is high angle whether supposed reversed or normal; in contrast, hypothesis C requires relative movements of the order of 100 km. It is probably fair to say that the geological evidence, as it stands at present, does not provide a generally acceptable answer.

Some help has been provided by gravity and aeromagnetic observations [*Girdler*, 1958], but such results provide evidence about the present situation only. A more important need is for a historical study by means of which the magnitude of the past relative movements between Africa and Arabia may be estimated. Paleomagnetism is a possibility. The Red Sea depression, and its continuation into the Jordan valley, pinches out northwards. Under the dilational hypothesis, therefore, a movement apart in the south of about 300 km is reduced to zero in Syria by an anticlockwise rotation of the Arabian peninsula about some point in Syria, and the paleomagnetic directions in rocks laid down before this movement took place should show systematic divergences in declination, but not in inclination, as between Africa and the Arabian peninsula. No divergences will be present if no such movement has occurred. A full paleomagnetic survey of

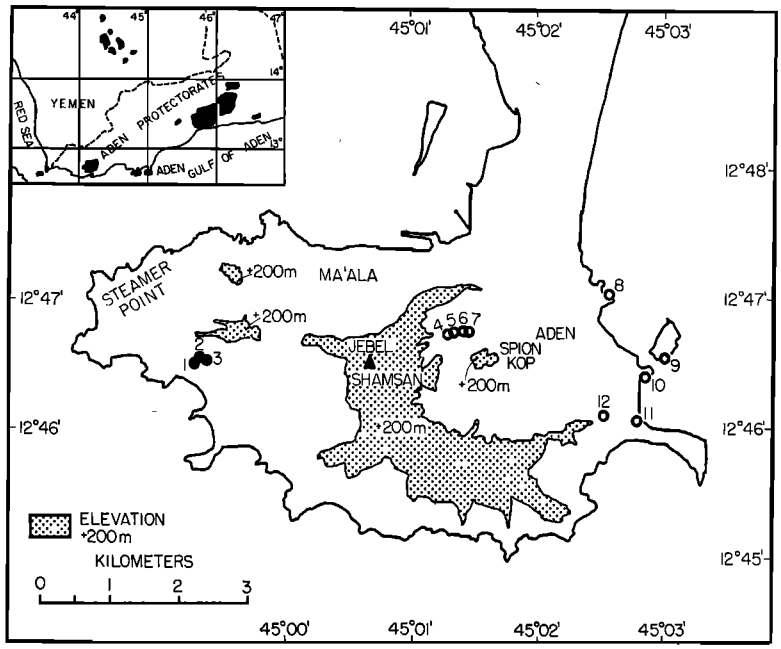


Fig. 1. Sketch-map of the Aden peninsula. The collecting sites are shown by dots (normal directions) and circles (reversed directions) and are numbered as in the text and in Table 1. The outcrop of the 'Younger Volcanics,' taken from *Wissmann, Rathjens, and Kosmat* [1942] are shown in solid black in the inset.

rock formations from both sides of the Red Sea will be necessary to complete this test.

2. Geology and Sampling

The volcanics of the Aden peninsula [*Mallet*, 1871; *MacMahon*, 1883; *Wissmann, Rathjens, and Kossmat*, 1942; *Jacques*, 1953–1954] belong to a group loosely termed the 'Younger Volcanics.' Five main areas of these stretch from Perim eastward into the Protectorate and northward into the Yemen (Fig. 1, inset). The relations of these volcanics to the underlying strata are usually obscured by desert sands, but where the contact is seen they rest unconformably on Jurassic and Cretaceous limestones and volcanics, the latter being referred to as the 'Older Volcanics.'

The Aden peninsula (Fig. 1) consists of a partly eroded volcano, tied to the mainland by a strip of Recent sand; the volcano consists of lava flows and pyroclastics intruded by trachyte plugs. The main crater, which contains the city of Aden, is in the eastern part, with Spion Kop as the central cone. The base of the Aden Volcanics is not exposed. Their age can only be estimated as Pleistocene or Recent from the extent of erosion and preservation of land forms, but there is no evidence to exclude an earlier age.

Sampling has been confined to twelve working quarries (sites numbered 1 to 12 in Fig. 1 and Table 1). Because of the extensive weathering of natural exposures it was not possible to obtain fresh samples from other places. Several separately orientated samples have been obtained from each site, 55 in all. One banded lava has been sampled at Ras Quatam (site 8). In the Hokat Bay and Khusaf Valley areas massive trachytes were sampled at eight sites (4 to 7, and 9 to 12). These appear to have the form of plugs, although it is possible that they are thick flows, since screes obscure the exposure on most sides. In the Gold Mohur area a trachyte plug has been sampled at three sites (1, 2, and 3). Here the intrusive nature is apparent, the adjacent bedding being upturned at the margins.

3. Results

Directions of natural remanent magnetization $\mathbf{M}_n$. Two to five cylindrical specimens have

21562202, 1961, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i002p00549 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1. *Natural Remanent Magnetization.* S is the number of samples and N the number of specimens cut from these. The site directions are specified by the declination D and inclination I reckoned clockwise from the north, R is the resultant of the N direction considered as unit vectors, and a is the error (P = 0.05) [*Fisher*, 1953]. $\bar{M}_n$ is the arithmetic mean of the intensity of natural remanent magnetization of specimens at each site in emu/cm³.

Site No.	Site Location		S	N	Site Direction D	I	R	a	$\bar{M}_n$
1	Gold Mohur Bay	12°46′30″N 44°59′15″E	5	15	354.4	+18.3	14.74	7.5	0.3
2	Gold Mohur Bay, 80 m W. of 1	12°46′32″N 44°59′18″E	5	15	343.2	+24.5	14.53	7.0	0.5
3	Gold Mohur Bay, 120 m S.E. of 2	12°46′31″N 44°59′20″E	5	15	353.0	+21.0	14.69	5.8	0.4
4	Khusaf Valley, 800 m W.S.W. of Old Residency	12°46′45″N 45°01′16″E	5	15	175.6	−22.2	14.77	4.9	1.2
5	Khusaf Valley, 200 m E. of 4	12°46′46″N 45°01′21″E	5	15	174.4	−18.4	14.53	7.2	1.4
6	Khusaf Valley 200 m E. of 5	12°46′46″N 45°01′24″E	5	15	179.1	−29.3	14.61	6.4	0.8
7	Khusaf Valley, 400 m S.W. of Old Residency	12°46′46″N 45°01′28″E	5	15	174.1	−25.0	14.86	3.8	0.2
8	Ras Quatam (banded lava)	12°47′02″N 45°02′32″E	2	10	157.9	−37.9	9.96	3.3	0.4
9	Sira Island	12°46′34″N 45°03′00″E	5	12	172.2	−22.8	11.83	5.4	0.2
10	Hokat Bay, north	12°46′27″N 45°02′50″E	6	15	177.5	−30.7	14.90	2.1	0.7
11	Hokat Bay, south	12°46′05″N 45°02′45″E	3	10	179.6	−16.7	9.80	5.0	0.7
12	Hokat Bay, Hindu Cemetery	12°46′08″N 45°02′29″E	3	10	173.9	−20.2	9.87	5.9	0.3

been cut from each sample (see Table 1 for details). The directions and intensity of magnetization have been measured by astatic magnetometers by the methods described by *Collinson, Creer, Irving, and Runcorn* [1957]. The directions in all 164 specimens are plotted in Figure 2. At sites 1 to 3 the directions are 'normal,' the inclinations are low and positive, and the declinations are mostly just west of true north. At sites 4 to 12 the directions are 'reversed,' the inclinations are low and negative, and the declinations are southerly, with a tendency to be east of south. At site 9 two specimens from the *same* sample have directions away from the main groups and from the other twelve specimens (from five samples) at site 9. This could be due to an orientation error, and the results from these two specimens are omitted from the analysis; however, their inclusion would not alter the results.

The mean directions at each site are listed in Table 1 and plotted in Figure 3. The mean directions of the normal and reversed groups are given in Table 2. The difference (4°) between these, irrespective of sign, is not significant. The overall mean direction, irrespective of sign, which is given in Table 2, is close to, but not exactly that of, the present geocentric axial dipole field (0, +24.4). The declination of the mean is N 6.9°W. The error (a) in the mean is 2.7°; the error in declination is therefore given by $a \sec I = 3.0°$, which is less than half of the divergence from true north. As a test of this analysis the horizontal component of the directions for the twelve sites may be summed, each site being assigned unit weight to ensure a conservative estimate. In this case $R = 11.95$; the 95 per cent error in the mean is therefore given by $(2 \cos^{-1}R/N)/N^{\frac{1}{2}}$, which is 3.8°. Alternatively, the frequency of occurrence of declination values may be analyzed, and, if sites are again used as units and a Gaussian distribu-

21562202, 1961, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i002p00549 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

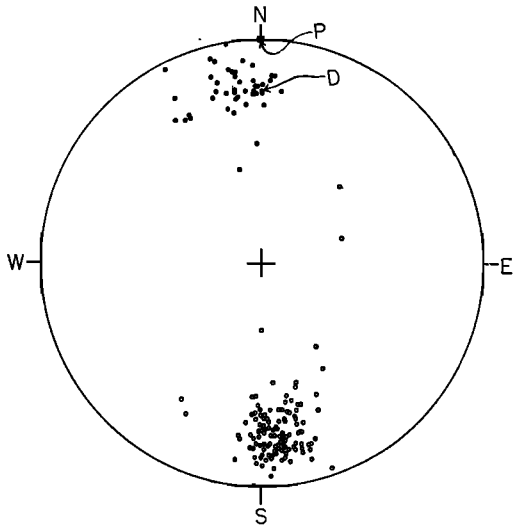


Fig. 2. Direction of natural remanent magnetization. The directions, M_n, in 164 specimens are shown on this equal-area net. The primitive is the present horizontal plane at Aden. North-seeking directions with positive inclinations are plotted as dots on the lower hemisphere and, when the inclinations are negative, by circles on the upper hemisphere. The present field directions (359.7, 0) and the geocentric dipole field directions (0, +24.4) are at the heads of the arrows labeled **P** and **D**, respectively.

tion is assumed, the 95 per cent error in the mean is 3.7°. These errors are somewhat greater than those obtained previously, as would be expected, because site mean values only have been used and the data are not so fully utilized. The measurements have been made with two magnetometers of different design, and the directions of magnetization obtained by these have been compared with each other and with instruments at the Universities of Adelaide and Cambridge [*Collinson, Creer, Irving, and Runcorn*, 1957]. We are satisfied, therefore, that this deviation of the overall mean direction from the present meridional direction is real. As a safe estimate we may give this divergence as 7 ± 4°W.

Intensity of magnetization. The mean values of the intensity of natural remanent magnetization M_n at each site are given in Table 1.

The maximum isothermal remanent magnetization $M_{i(\max)}$ and its coercivity H_{cr} have been measured by the method described by *Creer, Irving, and Nairn* [1959]. Typical values for one specimen with normal directions and for two with reversed directions—one (11) from massive trachyte and the other (8) from banded lava—are given in Table 3. The values of $M_{i(\max)}$ measured in eleven specimens from five sites range from 22 to 136, with an average of 70×10^{-3} emu/cm³. Values of H_{cr} range from 260 to 580 oersteds, with an average of 420. The fields required to saturate range from 2500 to 5000 oersteds.

The intensity of thermoremanent magnetization, $M_{t(0.6)}$, has been obtained by heating specimens in an atmosphere of nitrogen to 650°C and afterwards cooling them in the earth's field (0.6 oersted). The intensities are listed in Table 4. If the earth's field at Aden at the time the rocks cooled is assumed to have had approximately its present value (0.4 oersted), the original intensity $M_{t(0.4)}$ may be calculated, since M_t varies linearly with field in this range. The ratios, $M_n/M_{t(0.4)}$, given in Table 4 indicate

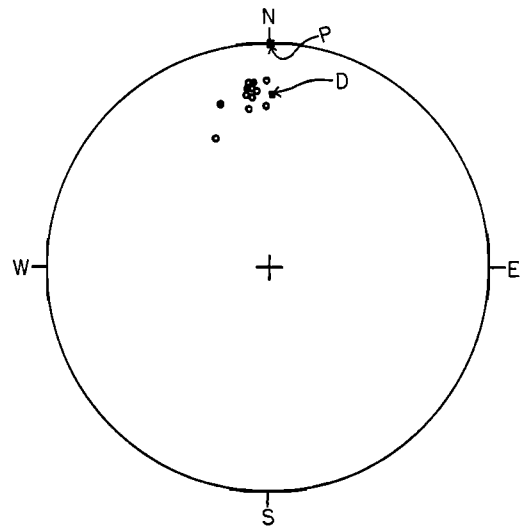


Fig. 3. Site directions. The projection is equal-area, and the primitive is the present horizontal. All directions are plotted on the lower hemisphere to show how nearly the normal and reversed sites are opposed to one another. Dots are north-seeking directions (sites 1 to 3) and circles are south-seeking directions (sites 4 to 12). The site mean directions are, of course, subject to errors (listed in Table 1), but to avoid confusion these are omitted here. The effect of these errors is eliminated in the analysis given in Table 2 so that the within-site dispersion ($w = 340$, circular standard deviation = 4.4°) is less than the dispersion (w') of the site directions ($w = 99$, circular standard deviation = 8.2°).

21582202, 1961, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i002p00549 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 2. Analysis of the directions of natural remanent magnetization. S is the number of samples and N is the number of specimens, D and I are the declination (reckoned clockwise from true north) and the inclination of the mean direction, w and b are estimates of the within- and between-site variation, k is an estimate of the over-all precision from which a, the error in the mean ($P = 0.05$), is calculated [*Fisher*, 1953; *Watson and Irving*, 1957].

Group	S	N	D	I	w	b	k	a
Normal (3 sites)	15	45	350.3	+21.3	40	299	549	6.0
Reversed (9 sites)	39	117	174.0	−24.9	58	369	2225	2 9
All, irrespective of sign	54	162	353.1	+24.0	52	340	2740	2.7

that a substantial proportion of the original thermoremanence remains.

The stability and origin of the natural remanent magnetization. The magnetization is not an isothermal remanent magnetization impressed by the present earth's field, since the individual specimen directions and the site mean directions differ significantly from this. Even a small component of this origin cannot be present because the opposed groups are exactly 180° apart.

Tests of stability have been made by treating specimens in 50-cps alternating magnetic fields; the apparatus described by *Irving, Stott, and Ward* [1960] was used in the tests. The procedure is to subject a specimen to a low alternating magnetic field and then to measure the direction and intensity of magnetization. Directions are measured to an accuracy of 3° and intensity is measured to 5 per cent. The treatment is repeated in a higher magnetic field, and measurements are made again, and so on. The peak values of field strength ($H_p\sim$) used were 38, 75, 112, 150, 225, 300, 450, 600, 750, and 900 oersteds. Specimens from eight sites have been treated in this way, and the changes in intensity for two specimens (sites 8 and 9) are given in Figure 4. The demagnetization curves for the other six specimens all fall between, or near to, these two curves. The intensity changes little in fields of up to 100 oersteds, but afterwards it falls off rapidly to a steady value which is always less than $0.5M_n$. These values are, in descending magnitude (site numbers in parentheses), 0.49 (11), 0.47 (8), 0.42 (1), 0.41 (5), 0.37 (3), 0.26 (12), 0.21 (9), and 0.16 (7) of M_n. During this treatment certain changes in direction occurred. These changes may be expressed as the circular standard deviation [*Creer, Irving, and Nairn*, 1959] of the direction changes for each specimen during the ten demagnetization steps. The deviations are (site numbers in parentheses) 1° (11), 2° (8), 1° (1), 2° (5), 1° (3), 19° (12), 2° (9), and 13° (7). In most cases these deviations are to be expected from the experimental error, and treatment in alternating fields of up to 900 oersteds does not affect the direction significantly. In only two cases are the measurement errors exceeded, and in these (specimens from sites 12 and 7) the means of the ten determinations did not differ significantly ($P = 0.05$) from the initial directions $\mathbf{M}_n$; therefore the changes produced by alternating fields are random. These changes are probably due to randomization by the soft components of the alternating fields, which are large in these particular specimens (see values of M/M_n given above and Table 4).

For comparison, the demagnetization curves of natural remanent magnetization M_n and thermoremanent magnetization $M_{t(0.6)}$ are similar to that for M_n, but the separation into two components is less clear, as if fusion had begun during laboratory heating. This is suggestive of exsolution intergrowths, with the fine-grained intergrowths contributing the hard

TABLE 3. $M_{i(\max)}$ is the saturation isothermal remanent magnetization, $H_{i(\max)}$ is the steady field required to saturate, and H_{cr} is the back field required to nullify $M_{i(\max)}$. The intensity of natural remanent magnetization, M_n, is given for comparison. Magnetizations are given in emu/cm^3 × 10^{-3} and fields in oersteds. Values are for the same specimen at each site.

Site	$M_{i(\max)}$	$H_{i(\max)}$	H_{cr}	M_n
2	27.8	2500	450	0.3
8	71.0	4000	360	0.4
11	60.5	5000	490	0.5

21562202, 1961, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i002p00549 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4. $M_{t(0.6)}$ is the thermoremanence acquired in a field of 0.6 oersted. M_n is the intensity of natural remanent magnetization, and the ratio $M_n/M_{t(0.4)}$ is an estimate of the proportion of original intensity remaining today. Intensities are given in emu/cm^3 $\times$ 10^{-3}. Values are for the same specimen at each site.

Site	$M_{t(0.6)}$	M_n	$M_n/M_{t(0.4)}$
11	0.85	0.47	0.8
10	1.90	0.96	0.7
4	3.06	1.14	0.6
7	1.33	0.24	0.3

component whereas the matrix, in which the crystal size is larger, contributes the soft component. In such a two-component system, the ratio of the hard component to the soft component is never likely to exceed unity but could have any value less than this, as is observed. Further evidence regarding the separation of the magnetization into two components is provided by the independence of thermoremanence and isothermal remanence in low fields. After demagnetization of $M_{t(0.6)}$ in a 900-oersted alternating field, the specimen was placed in a steady field of 300 oersteds and an isothermal magnetization $M_{i(300)}$ was induced. The demagnetization curve for $M_{i(300)}$ is quite different from that for M_n or $M_{t(0.6)}$. The moment ($M_{i(300)}$, disappears in an alternating field of 300 oersteds, and the direction and intensity remaining are exactly those present before $M_{i(300)}$ was induced. The hard component is unaffected by the acquisition or removal of an isothermal magnetization in 300 oersteds.

Both components are stable and have parallel directions, and the natural remanent magnetization is closely similar, both in its intensity (Table 4) and in its demagnetization characteristics (Fig. 5), to thermoremanence induced in the laboratory. There is no evidence that the rocks have been disturbed or altered since they were formed, and it is therefore reasonable to conclude that the natural remanent magnetization was acquired in the direction of the geomagnetic field at Aden during the time the rocks cooled.

4. Geomagnetic Aspects

Secular variation. Samples at each site were taken within a few meters of one another so that the site directions are spot readings of the geomagnetic field. The within-site dispersion w gives the estimates of the error of these readings (referred to as paleomagnetic errors), and the between-site variation b provides information about the secular variation during the range of time taken for the rocks at all sites to cool over the range of their Curie temperatures. Such information is important for discriminating between the models of the geomagnetic field assumed for the analysis of paleomagnetic data. Two models have so far been used. In both cases it is assumed that the main field has, on the average, been that of a geocentric dipole. The model proposed by *Creer, Irving, and Runcorn* [1954], which has been more generally used, implies that the site directions, if spread over a period of time longer than the period of secular variation, will have a Fisher distribution [*Fisher*, 1953] in which the directions will be symmetrical about the average or dipole field direction where the density is a maximum. More recently, *Creer, Irving, and Nairn* [1959] have proposed a second model for which it is assumed that the distribution will depend on the paleomagnetic colatitude ψ. It is not possible, from the existing data, to decide between these models; in any case, it is very probable that other models will eventually be necessary. At the present time we need very much more information about the distribution of site directions in Quaternary and Upper Tertiary rock formations in different latitudes.

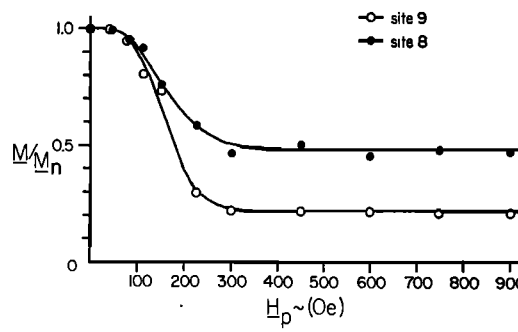


Fig. 4. Demagnetization of the natural remanent magnetization by alternating magnetic fields. The ratio of the intensity of magnetization (M) remaining after treatment to the intensity of the natural remanent magnetization (M_n), is plotted against the peak value of the applied field (H_p~). The upper curve is from a specimen from site 8 and the lower curve from a specimen from site 9.

21562202, 1961, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i002p00549 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

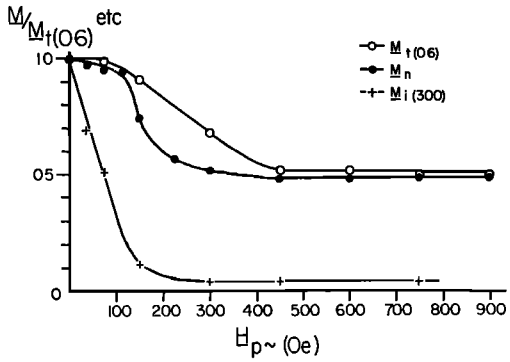


Fig. 5. Comparison of demagnetization of isothermal, natural, and thermoremanent magnetization in the same specimen from site 11.

Because of weathering, the number of sites at Aden was limited to twelve—too small for statistical fitting—but the magnitude of the between-site dispersion (Table 2, Fig. 3) is of interest. The equations describing the second model need not be repeated here. It is sufficient to say that they require that

$$\partial I_{(\psi=90)} = \delta D_{(\psi=90)} = \tfrac{1}{2}\delta D_{(\psi=0)} = \tfrac{1}{4}\delta I_{(\psi=0)},$$

where δD and δI are the divergences from the declination D and the inclination I of the mean direction to be expected at any given probability; therefore the equatorial distribution has an area 8 times that in polar latitudes. Estimates of the between-site dispersion b, freed from the paleomagnetic errors w, have not usually been given in the literature, only three such estimates being available: the Tasmanian Dolerite Sills, $b = 86$, $\psi = 10°$ [*Watson and Irving*, 1957]; Older Volcanics of Victoria, $b = 35$, $\psi = 31°$ [*Irving and Green*, 1957]; Newer Volcanics of Victoria, $b = 37$, $\psi = 49°$. Now in the Aden Volcanics ($\psi = 77°$), $b = 340$, the between-site dispersion is very much less than that in these rock formations and less than the present-day secular variation in high latitudes. The Aden results suggest that the secular variation at low latitudes during the time of cooling was very small, and in this feature they are inconsistent with the second model [*Creer, Irving, and Nairn*, 1959]. It is possible that the full range of the secular variation is not represented, but the presence of reversals suggests that this is not so.

Reversals. The mean of the two groups (Table 2) of directions are, within the errors, exactly 180° apart. This is the first known case of exact reversal in such low latitudes. Specimens at one site always have the same polarity, and the reversed sites are grouped to the east of the peninsula whereas the normal sites are to the west of it (Fig. 1). The reversals could be due to a reversal of the main earth's field or to a process of self-reversal within specimens, but the directions of thermoremanent magnetization ($\mathbf{M}_{t(0.6)}$) induced in the laboratory are always those of the applied field. The two magnetic components present (section 3) could provide a situation favorable to self-reversal if they had antiparallel directions, but they are always parallel in direction and, moreover, are found in all the specimens studied, whether normal or reversed.

The occurrence of reversals, assuming they are due to a field reversal, may assist in dating the Aden Volcanics more precisely. The last reversal in other parts of the world [see, for instance, *Roche*, 1956] occurred in the early part of the Pleistocene. It would appear that the reversed rocks of Aden are either early Pleistocene or older.

5. Tectonic Aspects

The inclination of the mean direction is the same as that of the geocentric axial dipole field (0, +24.4), but the declination is N 7° ± 4°W. This could be due to the rotation, subsequent to the Aden volcanism, of the Arabian land mass about a point in the region of what is now the Syrian desert. The amount and sense of the declination is right for closure of the Red Sea, assuming Africa to be unmoved. In Figure 6*a* the present coastlines and the meridian (45°E) through Aden are shown. The arrowed direction is the meridian obtained from the paleomagnetic results. In Figure 6*b* the Arabian peninsula is moved first by a rotation about a point in southern Syria and then translated southward approximately 150 km so that the opposite coastlines of the Red Sea fit closely to each other. The paleomagnetic meridian, carried with Arabia, is now parallel to the geographical meridian. The southwestern tip of the peninsula overrides a portion of Africa consisting of very young rocks which are thought to have been formed since the movement. Thus the paleomagnetic results are consistent with this movement, which is similar to that required by *Carey* [1958],

21582202, 1961, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i002p00549 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

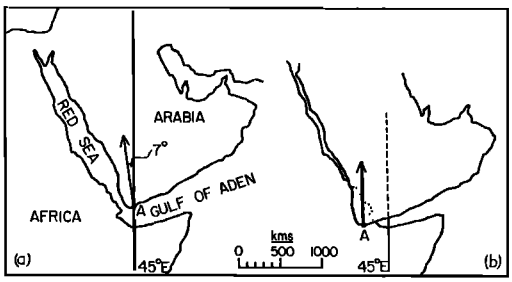


Fig. 6. 'Closure' of the Red Sea. The westerly declination observed in the Aden volcanics is of the right amount for the rotation of Arabia and closure of the Red Sea. Figure 6*a* shows the present coast lines; Aden is marked *A* and the arrow is the paleomagnetic meridian. In Figure 6*b* the Red Sea is closed and the paleomagnetic meridian is now parallel to the meridian fixed to Africa, which is assumed to be unmoved.

but they do not, of course, prove that it did occur. If this interpretation is correct, the Aden Volcanics were formed earlier than the Red Sea was formed, and a determination of their age by radiogenic decay would set a lower limit to the date of the Red Sea formation.

A second possibility is that the main field itself has moved relative to the earth as a whole, so that when these rocks were magnetized the northern-hemisphere pole of the dipole axis had the coordinates 83.2°N, 50.1°W (errors 2.8, 1.6 ($P = 0.05$)). This pole and the present geographic pole lie on the same small circle, with Aden at the center. A comparable result has been obtained in Japan [*Nagata, Akimoto, Uyeda, Shimuzi, Ozima, and Kobayashi*, 1957], where the mean pole (76.4°N, 37.6°E; errors 10.4, 7.1) for Pleistocene and Recent lavas lies on the same small circle as the geographical pole, with Japan at the center. This pole differs by a significant amount (17°) from that obtained from Aden. The Japanese result could be explained (like that from Aden) by an anticlockwise rotation of Japan by 16 ± 8° [*Irving*, 1959], and this is not unreasonable because Japan is a tectonically active region. These, however, appear to be special conditions, since in stable regions (Europe and Australia, for instance), where departures from the present pole have been recorded, both inclination and declination are affected. A third possibility is that Aden itself has rotated relative to Arabia.

To decide between these possibilities it will be necessary to establish whether or not systematic differences occur between rock formations in northeast Africa and in the Arabian peninsula. However, these first results make it clear that the paleomagnetic method is sensitive enough to provide an answer when results are available from both sides of the Red Sea.

References

Carey, S. W., A tectonic approach to continental drift, *Continental Drift: A Symposium*, Hobart, 375 pp., 1958.

Creer, K. M., E. Irving, and S. K. Runcorn, The direction of the geomagnetic field in remote epochs in Great Britain, *J. Geomag. Geoelec.*, *6*, 163–168, 1954.

Creer, K. M., E. Irving, and A. E. M. Nairn, Palaeomagnetism of the Great Whin Sill, *Geophys. J.*, *2*, 306–323, 1959.

Collinson, D. W., K. M. Creer, E. Irving, and S. K. Runcorn, The measurement of the permanent magnetization of rocks, *Phil. Trans. Roy. Soc. London A.*, *250*, 73–82, 1957.

Fisher, R. A., Dispersion on a sphere, *Proc. Roy. Soc. London, A*, *217*, 295–305, 1953.

Girdler, R. W., The relationship of the Red Sea to the East African rift system, *Quart. J. Geol. Soc. London*, *114*, 79–105, 1958.

Irving, E., Palaeomagnetic pole positions: a survey and analysis, *Geophys. J.*, *2*, 51–79, 1959.

Irving, E., and R. Green, The palaeomagnetism of the Kainozoic Basalts of Victoria, *Monthly Notices Roy. Astron. Soc., Geophys. Suppl.*, *7*, 347–359, 1957.

Irving, E., P. M. Stott, and M. A. Ward, The demagnetization of igneous rocks by alternating magnetic fields, *Phil. Mag.*, *6*, 1961.

Jacques, E. H., Geology of the Aden Protectorate, *Port of Aden Annual*, 46–48, 1953–1954.

MacMahon, C. A., On the lavas of Aden, *Records Geol. Survey India*, *16*, 145–158, 1883.

Mallet, F. R., On the geological structure of the country near Aden, *Mem. Geol. Survey India*, *7*, 257–284, 1871.

Nagata, T., S. Akimoto, S. Uyeda, Y. Shimuzi, M. Ozima, and K. Kobayashi, Palaeomagnetic studies on a Quaternary volcanic region in Japan, *J. Geomag. Geoelec.*, *9*, 23–41, 1957.

Roche, A., Sur la date de la dernière inversion du champ magnétique terrestre, *Compt. rend.*, *243*, 812–814, 1956.

Watson, G. S., and E. Irving, Statistical methods in rock magnetism, *Monthly Notices Roy. Astron. Soc., Geophys. Suppl.*, *7*, 289–300, 1957.

Wissmann, H., C. Rathjens, and F. Kossmat, Beitrage zur Tektonik Arabiens, *Geol. Rundschau*, *33*, 221–353, 1942.

(Manuscript received August 13, 1960; revised October 27, 1960.)

21562202, 1961, 2, Downloaded from https://agupubs.onlinelibrary.wiley.com/doi/10.1029/JZ066i002p00549 by Curtin University Library, Wiley Online Library on [27/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License